
COMPREHENSIVE ANALYSIS OF CLASSIFICATION AND GENERATIVE MODELLING

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ABSTRACT

This paper presents optimized deep learning architectures for efficient image classification and generation under strict computational constraints. A modified convolutional network inspired by SqueezeNet is developed for classification, balancing parameter efficiency with representational power through custom Fire Modules. For generative modeling, a tailored Deep Convolutional Generative Adversarial Network (DCGAN) is designed with architectural enhancements to improve image fidelity. Both models demonstrate the potential of lightweight networks to achieve competitive performance in resource-limited environments.

Part 1: Classification

1 METHODOLOGY

The method involves using the AddNIST dataset to train a deep classification network inspired by the SqueezeNet architecture [4] but with key modifications to optimize for parameter efficiency and performance. The architecture integrates custom **Fire Modules**, which consist of a *squeeze* layer using 1×1 convolutions, followed by an *expand* layer that combines both 1×1 and 3×3 convolutions. Unlike traditional SqueezeNet, which aggressively reduces the number of parameters through extensive use of 1×1 filters, our model balances parameter efficiency and representational power by slightly increasing the number of filters in each layer.

Key architectural choices include:

- **Initial Convolution Layer:** 55 filters of size 3×3 with stride 1, followed by batch normalization.
- **Fire Modules:**
 - Fire1: Squeeze (27 filters), Expand (1×1 and 3×3 convolutions with 55 filters total).
 - Fire2: Squeeze (29 filters), Expand with 56 filters.
 - Fire3: Squeeze (34 filters), Expand with 70 filters.
 - Fire4: Squeeze (35 filters), Expand with 70 filters.
- **Pooling Strategy:** Average pooling (AvgPool2d) is applied after each Fire module instead of max pooling, contributing to smoother feature extraction.
- **Activation Function:** The Exponential Linear Unit (ELU) $\varphi(x) = \begin{cases} x & \text{if } x > 0 \\ \alpha(e^x - 1) & \text{if } x \leq 0 \end{cases}$ is used for all convolutional layers, improving convergence and mitigating vanishing gradients [2].
- **Regularization:** A dropout rate of 0.34 is applied before the final classification layer to reduce overfitting.
- **Final Layers:** A 1×1 convolutional layer maps features to the 20 output classes, followed by global average pooling to reduce spatial dimensions.

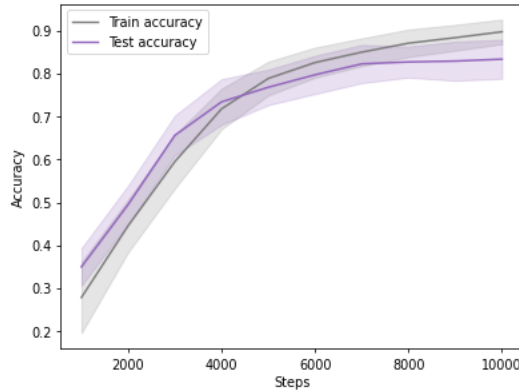
Hyperparameter tuning was performed using Optuna [1], optimizing the learning rate, dropout rate, activation function, and pooling type. The optimal configuration found includes a learning rate of 0.0035, ELU activation, average pooling, and a dropout rate of 0.34.

2 RESULTS

The network has 99,124 parameters, adhering to the constraint of fewer than 100,000 parameters.

It attains 89.7% training accuracy and 83.4% testing accuracy at 10,000 optimisation steps (train loss: 0.320, train acc: 0.897 ± 0.029 , test acc: 0.834 ± 0.045). These results demonstrate strong generalization within the allowed parameter and training step constraints.

Here is the training graph:



3 LIMITATIONS

While the model achieves high accuracy within the imposed constraints, further improvements could be made by experimenting with more advanced regularization techniques or deeper architectures. Although alternative activation functions such as LeakyReLU were explored during hyperparameter tuning with Optuna, the Exponential Linear Unit (ELU) consistently yielded superior performance, suggesting limited gains from switching activation functions. Similarly, pooling strategies were optimized, with average pooling outperforming other methods. Future work could focus on optimizing the network’s depth, introducing residual connections, or exploring more sophisticated data augmentation techniques to further enhance performance without exceeding the parameter budget.

Part 2: Generative Model

4 METHODOLOGY

The generative model is based on a modified Deep Convolutional Generative Adversarial Network (DCGAN) architecture [6], tailored to comply with the constraints of the assignment while optimizing for performance on the CIFAR-100 dataset [5]. This architecture maintains the core principles of DCGANs, leveraging convolutional layers for the discriminator and transposed convolutional layers for the generator, but introduces several key modifications for enhanced output quality and training stability.

The model consists of two networks:

- **Generator (G):** Transforms random noise vectors into synthetic images that mimic the visual characteristics of the CIFAR-100 dataset.

- **Discriminator (D):** Evaluates images to distinguish between real (from CIFAR-100) and generated images, providing adversarial feedback to improve the generator.

Key Modifications and Justifications:

1. **Enhanced Generator Architecture:** The generator employs four transposed convolutional layers with progressively decreasing channel dimensions: 192, 96, 48, and 3 (RGB output). This depth and filter configuration surpass the typical DCGAN, allowing for richer feature representation. Each layer is followed by batch normalization to stabilize training and **ReLU** activations to promote non-linearity, except the final layer, which uses a **Tanh** activation to scale outputs between $[-1, 1]$.
2. **Refined Discriminator Design:** The discriminator features four convolutional layers with increasing channel depth: 32, 64, 128, and 1. Unlike the standard DCGAN, the discriminator includes more aggressive batch normalization after each convolution to regularize learning. **LeakyReLU** activations with a negative slope of 0.2 are used to mitigate the vanishing gradient problem, enhancing gradient flow through the network.
3. **Hyperparameter Optimization with Optuna:** To achieve optimal training dynamics, hyperparameters were tuned using the Optuna framework [1]. The best configuration identified includes a learning rate of 0.0004 and a momentum term $\beta_1 = 0.7$, diverging from the typical DCGAN value of $\beta_1 = 0.5$. This adjustment improved the balance between generator and discriminator convergence rates, reducing training instability.
4. **Loss Function:** Both networks were trained using Binary Cross-Entropy Loss (BCE). The generator minimizes the loss by attempting to maximize the probability of the discriminator classifying generated images as real, while the discriminator minimizes the loss by correctly distinguishing between real and synthetic images. The adversarial loss function is expressed as:

$$\mathcal{L}_{\text{GAN}} = \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_z} [\log(1 - D(G(\mathbf{z})))] \quad (1)$$

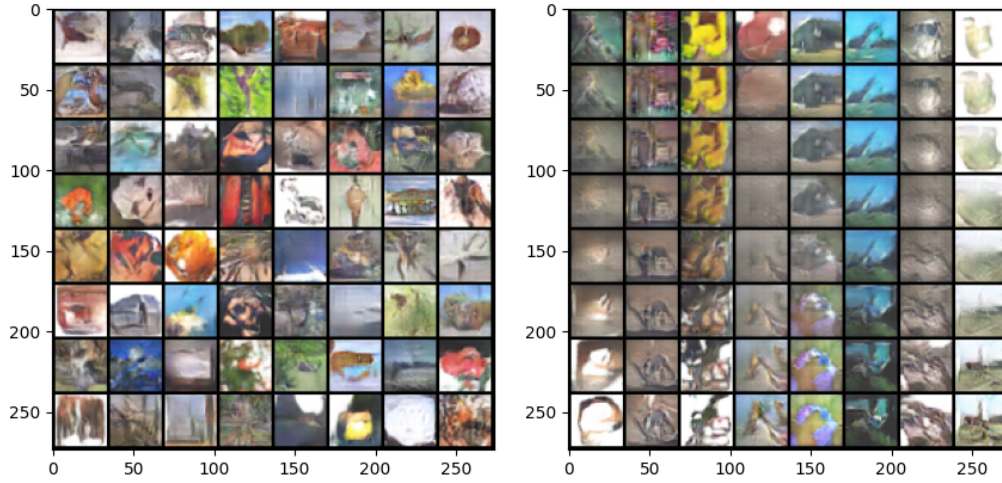
5. **Latent Space and Sampling:** A 100-dimensional latent vector $\mathbf{z} \sim \mathcal{N}(0, 1)$ is sampled from a Gaussian distribution. This vector is then upsampled through the generator to produce a 32×32 RGB image. The interpolation between latent vectors in the latent space demonstrates the continuity and generalization capacity of the generator.

5 RESULTS

The combined generator and discriminator model consists of **846,624 parameters**, comfortably within the 1,000,000-parameter limit. Training was conducted for **50,000 optimization steps**, adhering strictly to the computational constraints imposed by the assignment.

The model achieved a Fréchet Inception Distance (FID) score of 55.58 when evaluated against the CIFAR-100 test dataset [3]. The FID score measures the similarity between the generated images and real images from the dataset, with lower scores indicating higher visual fidelity and diversity.

A batch of 64 non-cherry-picked generated samples and latent space interpolations are presented below, demonstrating both the diversity and smooth transitions in the latent space:



The perceptual similarity of generated images was evaluated using the LPIPS metric [7], which measures deep feature space distance. Additionally, cherry-picked samples highlighting the most visually realistic outputs generated by the model are provided for reference:



6 LIMITATIONS

While the model demonstrates reasonable diversity and structure in its generated images, there are several limitations that hinder its performance. The images tend to exhibit blurriness and lack fine-grained details, particularly in complex objects. The relatively high FID score suggests that while the model captures the general structure of CIFAR-100 images, it struggles with generating sharp textures and realistic edges.

Another limitation observed is partial mode collapse, where certain classes or image structures dominate the generated outputs, reducing overall diversity. This indicates that while the generator has learned to produce certain types of images effectively, it has not generalized across the full distribution of the CIFAR-100 dataset. Further refinement in the balance between generator and discriminator training dynamics could potentially mitigate this issue.

Lastly, despite the optimized hyperparameters, training instability was occasionally observed, requiring careful monitoring to prevent divergence. Future work could explore deeper architectural tuning within the current framework to further stabilize and improve the generation quality without exceeding the parameter constraints.

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